Alejandro “Alex” Gonzalez

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Linkedin | Github | Portfolio

Education

BSc in Data Science, Minor in Computer Science, **Expected 2026**
Saint Mary's College of California, Moraga, CA, USA

Relevant Courses: Data Structures and Algorithms, Natural Language Processes, Statistics and Probability, Advanced Quantitative Analysis, Machine Learning, Data Warehousing, Web Development, Data Warehousing, Data Science Capstone

Academic Projects

Probabilistic Modeling & ML Systems **October 2024 – Present**
HoopVista: Prop Predictor (Link)

- Built HoopVista, an end-to-end NBA player prop prediction system, to identify positive expected value bets and DFS entries by replacing manual research with a data-driven model; the system scrapes live bookmaker lines, generates daily slates across PrizePicks, Underdog, DraftKings Pick6, and Betr DFS, and serves projections through a custom web interface
- Engineered a quantile regression pipeline (XGBoost, Q10/Q50/Q90) across four sub-models — minutes, points, assists, and rebounds per minute — trained with walk-forward time-series validation to prevent leakage, with separate model tiers for stars vs. role players to counter regression-to-the-mean bias, and Bayesian context adjustments for injuries, pace, spread, and defensive matchups
- Implemented a Monte Carlo simulation layer (10,000 trials) with sharp-book cross-referencing to compute probability for over and under per DFS line, tiering each pick against Pinnacle/FanDuel consensus odds, and assembling optimal 2-, 3-, 5-, and 6-leg entries using expected value (EV) to filter them

Markov Chain Monte Carlo **April 2026 – May 2026**
Heatcheck: NBA Points Simulator (Link)

- Built HeatCheck, a Markov chain Monte Carlo simulator for NBA player scoring props, motivated by the lack of publicly available tools that model within-game performance momentum rather than relying on simple career averages
- Engineered a weighted state-transition model in Python that classifies each quarter of a player's career into four performance states (Cold → On-Fire) using personal scoring percentiles, then builds context-adjusted transition matrices across 3 seasons of play-by-play data to capture how a player's performance evolves quarter-to-quarter
- Deployed a Streamlit web app that runs 10,000 Monte Carlo simulations per query — incorporating opponent defensive rating, game pace, and spread as similarity weights — to output calibrated over/under probabilities and a full projected scoring distribution

Skills & Interests

Languages: Python, SQL (PostgresSQL), HTML/CSS/JS, Java, R
Machine Learning: Supervised/Unsupervised Learning, NLP
Libraries & Frameworks: Matplotlib, Pandas, NumPy, Seaborn, PyTorch, TensorFlow, scikit-learn
Tools: Git, Tableau, Microsoft Excel, PowerPoint, MS Outlook, Databricks